Predicting First-Year Student Enrollment Yield Using Admissions Data at NC State

CSC 522 – Group 5

Sparsh Koyambreth
skoyamb@ncsu.edu

Eshwar Kanikanti
ekkanika@ncsu.edu

1 PROBLEM STATEMENT

Every year, thousands of students are accepted into North Carolina State University to pursue higher education opportunities across over 100 different prospective majors. However, despite being accepted, many students choose to attend different schools. We would like to create a model based on demographic, academic, application, and financial age features to predict whether or not a student will accept their admissions offer to optimize recruitment spending.

2 RELATED WORK

A student's academic decisions and preferences are crucial for securing a career and knowing what they want to do in the future. The first year of college is the time when students make the most significant life decisions. The college they attend and the field of study they choose will shape the rest of their lives. An article written by Flora Zyberaj et al. describes a study aimed at evaluating psycho-emotional, social, and societal factors that influence students' decisions and university preferences and their self-reports based on their decisions [17]. A sample of 356 students were chosen, majority of whom were female and around 20 years old, and each were given a questionnaire with four groups of factors with demographic data to determine how important those factors affected their academic decisions. They found that students often rely on intuitive judgments and emotional responses, rather than rational analysis, when making academic decisions. Family is also a primary factor as parents usually pay for their children's education, and they have a huge affect on the selection of students' majors. The results of the study were that social and emotional factors have the greatest impact on students' decisions and that counseling services should be made available to students in order to give them more information and help them make better decisions regarding their career and job search.

In another article written by Nur Laila Ab Ghani, Zaihisma Che Cob, Sulfeeza Drus, and Hidayah Sulaiman, they demonstrated how to use Data Mining to predict and applicant's enrollment decisions in higher education institutions [8]. They used the logistic regression, decision tree, and naive bayes predictive models on a data set and tested them for accuracy using 10-fold cross validation. They categorized the data into different clusters based on personal, financial, and institutional factors, each of which contained sub factors such as gender, religion, tuition cost, course structure, location, and family income. The models were able to predict students' enrollment decision with around 70% accuracy with the decision tree model yielding the highest accuracy. The study concluded that personal and parental factors are the most important factors for determining a student's enrollment decision, with the top three being anticipated major, parent's employment status, and type of application.

Logistic regression is a very useful supervised learning algorithm used for classification problems and for predicting the probability of events. In a study by Gunaratnam Bakeerathan, he researched how to use logistic regression to identify factors and predict student enrollment decisions [2]. The choice model was validated with enrollment data using a classification table with 71.4% accuracy. The results of the model were that student GPA, LSAT score, scholarships, and distance to campus were all important factors in determining student enrollment. It was also decided that the model should be re-evaluated and updated each year since enrollment patterns might change. Logistic regression worked best for this specific study, but it is important to test between multiple models and methods as some might work better than others depending on the scenario.

3 PROPOSED APPROACH

3.1 Novel Aspects

This project introduces multiple novel features that distinguish it from conventional machine learning tasks. We intend to synthesize a dataset to answer the problem statement and identify key trends in the sample space, so that NC State Admissions can extract meaningful information.

3.1.1 Tailoring to NC State. A key part of what makes this project unique is its direct relevance to NC State University's admissions operations. Unlike generic Kaggle datasets that lack a real-world context, our dataset is carefully curated using actual distributions from NC State Common Data Set (CDS) 2024-2025 [13]. This makes sure that our findings are reflective of the true demographic, academic, and financial characteristics of the admitted applicants from this application cycle.

3.1.2 Synthetic Data Generation with Statistical Fidelity. Our approach for synthesizing data preserves the complex statistical relationships present in real admissions data. This result was achieved by:

(1) Maintaining exact marginal distributions from the CDS for over 20 variables including demographics, test scores, GPA, and financial aid characteristics.
(2) Modeling realistic conditional dependencies such as the relationship between residency status and financial need, or between intended major and test score submission rates.
(3) Calibrating enrollment probabilities using domain knowledge about the potential factors influencing student yield, including residency status (in-state students yield 45% compare to out-of-state at 25%), application timing (early action

applications yield 35% higher), and financial aid leverage from student packages.

3.2 Method

Our method includes the process of curating the dataset, building the models, and performing a comprehensive analysis, ensuring best practices are enforced while addressing any potential challenges to admissions data.

3.2.1 Dataset Curation. The initial step is to synthesize a dataset using weighted randomness based off of the proportions from the NC State Common Data Set (CDS), depicted in Section 4.

3.2.2 Model Building. After research, we figured on using the Decision Tree, Random Forest and Support Vector Machine (SVM) models, and compare them together using set benchmarks to predict if a student is going to enroll at NC State after getting admitted. We trained and evaluated these three classification algorithms because they each represent different approaches covered in CSC 522.

We implemented a decision tree using the Gini impurity criterion [4]. Decision trees recursively partition the feature space by selecting splits that maximize information gain, creating a structure of decision rules. Hyperparameters are then tuned via 5-fold cross-validation included maximum tree depth $\{5, 10, 15, 20, \text{None}\}$, minimum samples per split $\{20, 50, 100\}$, and minimum samples per leaf $\{10, 20, 50\}$. The primary advantage of decision trees is how simple they are to interpret. Admissions officers can trace exact decision paths and understand why specific students receive certain predictions. However, single trees are prone to overfitting and may have high variance, meaning small changes in training data can produce substantially different models [9].

To address the instability of single decision trees, we utilized the Random Forest model, an ensemble method that combines multiple trees through bootstrap aggregating (bagging) with random feature selection [3]. Each tree in the forest is trained on a bootstrapped sample of the data, and at each split, only a random subset of features is considered. This randomization decorrelates the trees, reducing variance and bias [12]. We tuned the number of estimators $\{100, 200, 500\}$, maximum features per split $\{\sqrt{p}, \log_2(p)\}$ where p is the total number of features, and maximum depth $\{10, 20, \text{None}\}$. Random forests typically achieve better predictive accuracy compared to single trees and provide robust feature importance rankings through mean decrease in impurity [9].

Finally, we implemented an SVM with a Radial Basis Function (RBF) kernel to capture non-linear decision boundaries [6]. The RBF kernel maps the original feature space into a higher-dimensional space where linear separation becomes possible:

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2) \quad (1)$$

where γ controls the influence radius of individual training samples. We tuned the regularization parameter $C \in \{0.1, 1, 10, 100\}$, which controls the trade-off between maximizing the margin and minimizing classification error, and the kernel coefficient γ. Class weights were set to 'balanced' to handle the imbalanced enrollment distribution (32.1% enrolled vs. 67.9% not enrolled). While SVMs excel at finding complex decision boundaries and are effective in high-dimensional spaces, they are computationally intensive for large datasets and lack ease of interpretation that tree-based methods have [9].

3.3 Rationale

Our methodology reflects careful consideration of the student population, course material, and practical constraints relative to this problem. We used synthetic data because real admissions data contains personally identifiable information (PII) and is protected by FERPA[1] regulations, making it unethical to use for this project. Our method of data generation allows us to work with realistic statistical distributions without privacy concerns, control the ground truth enrollment probabilities for validation, generate additional samples for testing, and share the dataset for reproducibility. The CDS provides us with publicly available average statistics that serve as reliable population parameters.

We selected Decision Trees, Random Forests, and SVMs because they represent a natural progression from simple to complex models while covering course topics. Decision trees are the foundation of tree-based methods, random forests demonstrate ensemble learning principles, and SVMs represent kernel methods for non-linear classification, specifically with RBF kernels to handle when the decision boundary may be complex.

Furthermore, focusing on enrollment yield prediction has significant practical value for the university. They can utilize this information to improve budget planning for the upcoming year, as a large part of a universities revenue comes from tuition payments from students. Housing is also very limited, as residence halls need to plan months in advance and this model can note when they reach headcount. Similarly, departments could also use enrollment forecasts to estimate how many students will be in certain classes which can alert them if they need to hire new faculty to not spread resources too thin.

4 DATASET

For this project, we synthesized our own dataset of exactly 18,201 students based off of the distributions from the CDS. This section describes the dataset structure, generation process, and statistical properties used.

4.1 Dataset Overview

The dataset simulates NC State's Fall 2024 admitted student pool and contains 18,201 observations, one row per admitted student to match the actual admitted count from CDS p.13. There are 22 variables, 1 being the student ID, 20 predictor features, and 1 binary target variable. There were 5,837 students in the positive class (accepted offer to attend NC State) and 12,364 students in the negative class (rejected offer to attend NC State). A preview of the dataset is shown in Table 1.

The data generation process is implemented in Python using NumPy for random sampling, Pandas for data manipulation, and is fully reproducible with fixed random seed. The complete synthesis code is available in our GitHub repository[2].

[1]Family Educational Rights and Privacy Act of 1974

[2]https://github.com/skoyamb_ncstate/CSC522_Final_Project/tree/skoyamb/dataset-gen

Table 1: The first 10 rows of the student enrollment dataset.

student_id	gender	residency	race_ethnicity	high_school_gpa	...	total_aid_package	enrolled
NCSU000001	Male	Out-of-State	White	4.000	...	0.00	0
NCSU000002	Female	International	White	3.904	...	0.00	0
NCSU000003	Female	Out-of-State	White	4.000	...	0.00	0
NCSU000004	Female	In-State	Asian	3.871	...	10939.31	1
NCSU000005	Male	In-State	White	3.562	...	0.00	0
NCSU000006	Male	In-State	White	4.000	...	0.00	1
NCSU000007	Male	Out-of-State	Hispanic/Latino	3.855	...	0.00	0
NCSU000008	Female	In-State	Black/African American	3.873	...	11758.53	1
NCSU000009	Female	Out-of-State	White	3.552	...	20565.07	0
NCSU000010	Female	In-State	White	3.805	...	0.00	1

4.2 Feature Descriptions

4.2.1 Demographic Features. The gender distribution follows CDS p.13 with 43.8% male and 56.2% female. Residency status reflects CDS p.14 with 58.1% in-state, 38.2% out-of-state, and 3.7% international students. Race/ethnicity follows eight categories from CDS p.6, with White (66.2%), Hispanic/Latino (9.9%), Asian (9.5%), and Black/African American (5.5%) which constitute the majority. First-generation college student status, estimated at 20% prevalence, is included as research shows first-generation students exhibit different enrollment patterns [14].

4.2.2 Academic Features. High school GPA follows the exact distribution from CDS p.21 with 20.0% with a 4.0 GPA, 56.7% in the 3.75-3.99 range, and 19.4% in the 3.50-3.74 range, resulting in a dataset mean of 3.83. Class rank categories (top 10%, top 25%, top 50%, bottom 50%) align with CDS percentile data as well. For standardized test scores, we model NC State's test-optional policy with 24.9% of students submitting SAT scores (25th percentile is a 1290, 75th percentile is a 1440) and 41.3% submitting ACT scores (25th percentile is 25, 75th percentile: 32) per CDS p.19-20. There is a binary indicator present for whether or not a student submitted their test scores.

4.2.3 Application Features. Application type distinguishes between Early Action (Nov. 1 deadline) and Regular Decision applications, with around 40% of students applying Early Action based on typical selective university patterns. Research indicated early applications demonstrate higher yield due to a higher level of interest [1]. Intended major follows degree conferral distributions from CDS p.41-42, with Engineering (24%), Business (17.5%), Biological Sciences (10.1%), and Computer Science (4.6%) as the most common student intended programs.

4.2.4 Financial Aid Features. Financial aid characteristics model the critical role of cost in enrollment decisions [10]. We track whether students applied for aid (60%), demonstrated financial need (33%), and received need-based aid (averaging $14,899 per CDS p.32). Merit scholarship recipients (15%) averaged $5,697 of aid per CDS p.34. Total aid package is the sum of need-based and merit scholarships, with a maximum package of $50,000.

4.3 Target Variable

The binary enrollment outcome ($Y \in \{0, 1\}$) was created using a logistic probability model tuned to reproduce realistic enrollment patterns observed in higher education research. The base enrollment probability for each student i is calculated by:

$$P(Y_i = 1|X_i) = \sigma\left(\sum_j \beta_j X_{ij}\right) \quad (2)$$

where $\sigma(z) = 1/(1 + e^{-z})$ is the logistic function and coefficients β_j reflect statistical findings on factors that drive enrollment. Residency effects were the strongest as in-state students received a 1.40× multiplier on base yield, while out-of-state (0.78×) and international students (0.65×) showed progressively lower enrollment rates, consistent with geographic proximity research [5]. Early Action applicants received a 1.35× boost reflecting higher demonstrated interest. Financial aid exerted substantial leverage, with aid package size providing up to a 2.0× multiplier for the largest packages, reflecting price sensitivity in enrollment [15].

Students in the top 10% of their class received a 0.85× multiplier, modeling the reality that the most competitive students often have multiple prestigious options and may choose higher-ranked institutions other than NC State. First-generation students received a 1.15× boost, reflecting research showing that first-generation students, once admitted, are more likely to enroll [16]. All multipliers were calibrated such that the dataset-wide mean enrollment probability matched exactly 32.1%, and enrollment decisions were sampled from these probabilities to generate the final binary outcomes.

5 EXPERIMENTS

5.1 Hypothesis

The main goal of the experiment was to see if by using our classifiers, can we create an effective model that can accurately predict whether a student is going to accept or deny their acceptance to NC State? Furthermore, we want to compare our models to see which one performs the best. We predict that financial aid will be the strongest predictor of enrollment, as prior research consistently identifies cost and financial aid as primary drivers of college choice, especially for students with financial need [10, 15]. We also expect that ensemble methods like Random Forest will outperform singular decision trees, as bagging fosters a higher predictive accuracy compared to a single decision tree [7].

5.2 Experimental Design

All the experiments performed followed a standardized protocol to make sure that there was a fair comparison to be made and that the process is reproducible, and is found in our GitHub repo.[3]

5.2.1 Data Preprocessing. Continuous features (GPA, test scores, financial aid amounts) were standardized using StandardScaler to achieve zero mean and unit variance, which is essential for SVM performance [11]. Categorical variables (gender, residency, race/ethnicity, application type, intended major) were one-hot encoded, creating binary indicator columns from these categories. Missing test scores (representing non-submission) were imputed with the median score among submitters for that test, with separate binary flags indicating submission status to store information about test-optional behavior.

5.2.2 Train-Test Split. The dataset was split 80-20 into training (14,561 students) and test sets (3,640 students) using stratified sampling to maintain the 32.1% enrollment rate in both partitions. The random state was fixed at 42 for reproducibility, similar to course assignments. The test set was held out completely during hyperparameter tuning and model training, ensuring unbiased performance evaluation.

5.2.3 Hyperparameter Tuning. Each model went through grid search with 5-fold cross-validation on the training set. Cross-validation splits were stratified by enrollment outcome to maintain class balance in each fold. The optimization metric was F1-score rather than accuracy, since the baseline accuracy of always predicting "not enrolled" would be 67.9%, which would provide no useful predictions for the enrolled class. F1-score balances precision (avoiding false predictions of enrollment) and recall (identifying students who will actually enroll), both critical for admissions planning.

5.2.4 Evaluation Metrics. Model performance was assessed using standard model evaluation metrics, such as accuracy, precision, recall, and F1-score. Accuracy tells us how often the model correctly guesses a student's final decision, whether they decided to enroll or not. Precision tells us out of all the students the model *claimed* were going to enroll, how many actually did? Recall states how often the model correctly predicted enrollment over the actual enrollment values. The F1-Score is the harmonic balance between precision and recall. The AUC is the probability that if you picked one random student who actually enrolled, and one random student who did not, the model gives a higher enrollment probability to the actual enrollee. Finally, specificity measures out of all the students who *actually declined* their acceptance to NC State, how many did the model successfully identify as not enrolling? Equations for these are stated below, utilizing dataset terminology.

$$Accuracy = \frac{\text{Correct Enroll + Correct Non-Enroll}}{\text{Total Students}} \tag{3}$$

$$Precision = \frac{\text{Correct Enroll}}{\text{Total Predicted Enroll}} \tag{4}$$

$$Recall = \frac{\text{Correct Enroll}}{\text{Total Actual Enroll}} \tag{5}$$

$$F1\text{-}Score = 2 \times \left(\frac{Precision \times Recall}{Precision + Recall}\right) \tag{6}$$

$$AUC = P(Prob_{Enrolled} > Prob_{Non\text{-}Enrolled}) \tag{7}$$

$$Specificity = \frac{\text{Correct Non-Enroll}}{\text{Total Actual Non-Enroll}} \tag{8}$$

5.2.5 Subgroup Analysis. To evaluate fairness and equity, we disaggregated performance metrics by residency status (in-state, out-of-state, international), application timing (Early Action vs. Regular Decision), and race/ethnicity (URM[4] vs. non-URM).

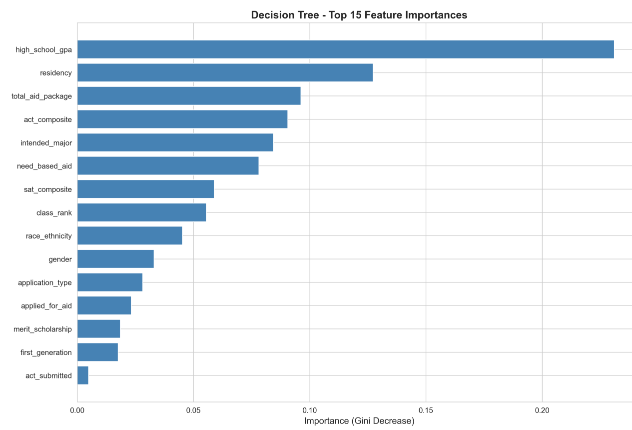


Figure 1: Most impactful features for the Decision Tree model.

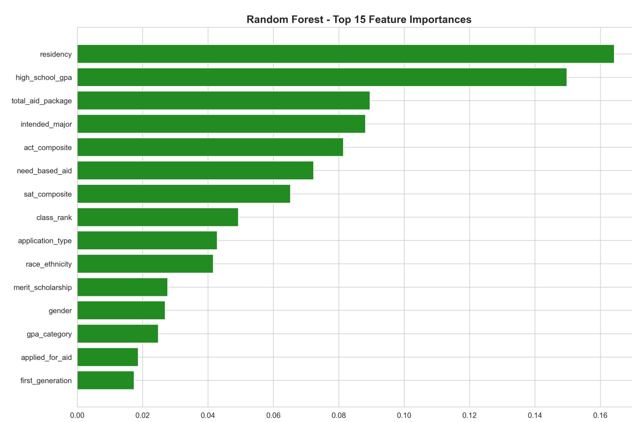
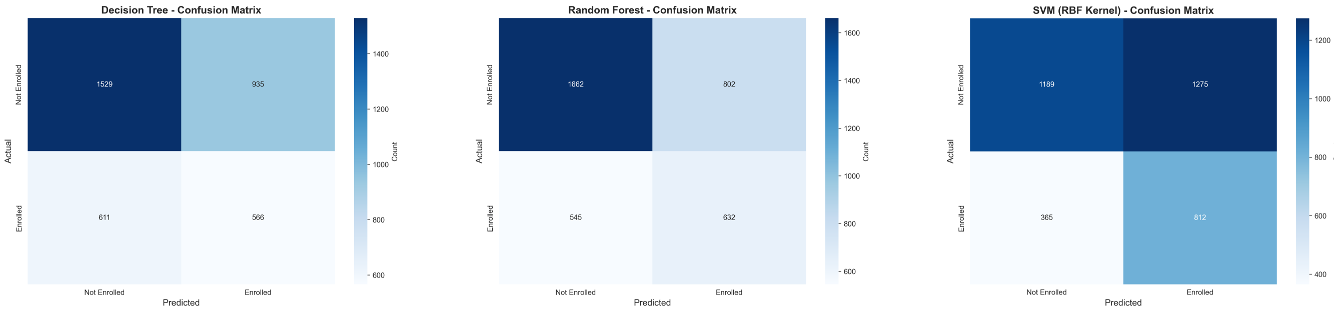


Figure 2: Most impactful features for the Random Forest model.

[3] https://github.com/skoyamb_ncstate/CSC522_Final_Project/tree/main/model

[4] Under-Represented Minority



(a) Decision Tree **(b) Random Forest** **(c) SVM (RBF Kernel)**

Figure 3: Confusion matrices evaluating the predicted vs. actual enrollment for each model.

6 RESULTS AND DISCUSSION

6.1 Model Performance

6.1.1 Class Imbalance Problem. After training our models, the respective performance based on the features described beforehand can be found in Figure 4. In terms of accuracy, the Random Forest model proved to be most effective in terms of the predicted class being correct. However, due to the class imbalance (roughly 67% did not enroll and roughly 33% enrolled), these models actually perform slightly worse than always guessing that a student will not enroll. For this reason, accuracy is not the most suitable metric for evaluating the performance of a model in this context.

6.1.2 Evaluation of Decision Tree. The decision tree represents the most simple model in our ecosystem. A visual representation of the splitting criterion is present in Figure 5, where the model attempts to segment the data based on features meeting a certain threshold defined. Residency proved to be the most effective split, starting from the root node which segments the data into in-state vs. out-of-state and international students. This is reasonable, as the cost of tuition ranging from $9,028 for in-state students, to $33,993 for out-of-state and international students[5], and the cost of attendance is one of the leading drivers behind a student's decision to attend a school [10].

This sentiment is corresponded by the feature importance histogram in Fig. 1, with residency status being the second-most impactful metric present. The most interesting aspect about the decision tree is that it valued an applicant's high school GPA significantly more than other features. This suggests that the decision tree identifies academic preparedness as the strongest singular splitting criterion, with students at different GPA levels exhibiting fundamentally different enrollment patterns. We believe this occurs because GPA serves as a proxy for competitiveness of alternative options as students with 4.0 GPAs likely have admission offers from more selective institutions and thus lower yield, while students with GPAs in the 3.5-3.75 range may view NC State as their strongest option.

6.1.3 Evaluation of Random Forest. The random forest represents the introduction of ensemble learning, generating many instances of decision trees and aggregating their predictions to mitigate the overfitting often seen in single decision trees. As seen in Fig. 4, the Random Forest achieved the highest overall Accuracy (0.6300), Precision (0.4407), ROC-AUC (0.6286), and Specificity (0.6745) among the three models. The strong Specificity score indicates that this model is particularly adept at correctly identifying students who will not enroll, which aligns with the underlying class imbalance of our dataset. Interestingly, when looking at the feature importance in Fig. 2, the Random Forest values residency slightly higher than high school GPA, flipping the top two features of the standalone Decision Tree. This indicates that when evaluating multiple decision paths and reducing variance, the geographic and financial implications of a student's residency become the most consistently dominant predictive factors. By considering a random subset of features at each split, the Random Forest ensures that a single dominant feature (like GPA) does not overshadow broader, systemic trends across the dataset.

6.1.4 Evaluation of SVM. The Support Vector Machine (SVM) represents the most complex mathematical model in our ecosystem. By utilizing a Radial Basis Function (RBF) kernel, the model attempts to project the applicant data into a higher-dimensional space to find a non-linear hyperplane separating enrolling and non-enrolling students. While it yielded the lowest overall Accuracy (0.5496) and Specificity (0.4825), Fig. 4 highlights that it significantly outperformed the other models in Recall (0.6899) and achieved the highest F1-score (0.4975). The confusion matrix in Fig. 3 illustrates this behavior, as the SVM is highly effective at capturing students who will actually enroll (812 True Positives, the highest of the three), effectively minimizing false negatives. However, it does so at the cost of misclassifying a much larger number of non-enrolling students as false positives. In the context of university admissions, this high-recall behavior might be desirable if the institution's goal is to cast a wide net and capture as many prospective enrollees as possible for targeted marketing, even if it means expending outreach resources on some students who ultimately decline their offers.

7 CONCLUSION

7.1 Lessons Learned

After completing this course project, we learned the significance of creating multiple models to evaluate a classification problem.

[5]https://studentservices.ncsu.edu/finances/estimated-cost-of-attendance

There is no "one-size-fits-all" model, and experimenting and tuning different hyperparameters is imperative to improving model performance.

We found that the "best" model for our dataset heavily depended on the specific evaluation metric prioritized. While the Random Forest achieved the highest overall accuracy and precision, the SVM with an RBF kernel proved superior in recall and F1-score, making it the most effective at identifying students who would actually enroll. However, because they rely on ensemble and kernel-based methods, these models are inherently more difficult to interpret than the simplistic Decision Tree. While the Decision Tree had lower overall performance, it provided clear, actionable insights into the driving factors of enrollment, such as residency and high school GPA. Choosing the right model ultimately requires balancing the need for raw predictive power against the need for transparency and interpretability, as simplicity may be preferred when you need to interact with non-technical users.

Another major outcome from this project was learning how to navigate real-world data challenges, particularly the class imbalance problem. Because our dataset heavily skewed towards non-enrolling students, we learned that relying solely on accuracy can be highly misleading. This prompted us to evaluate our models using a broader suite of metrics, allowing us to see the true strengths and weaknesses of each classifier. By combining these rigorous evaluation techniques with feature importance visualizations, we were able to build a comprehensive pipeline that not only predicted outcomes but also provided a deeper understanding of the underlying data.

7.2 Future Work

To improve the efficacy of these models, one immediate step is to consolidate multiple application cycles and aggregate the data together. Expanding the dataset across several academic years would not only provide a larger training set but also allow the models to capture broader trends, such as shifting economic factors, changes in university tuition policies, or evolving applicant demographics. This would lead to a noticeable increase of dataset size, increasing computational time for training, but lead to more actionable insights.

Additionally, future iterations should directly address the class imbalance problem at the data processing level. While we successfully adjusted our evaluation metrics to account for skewness, employing data augmentation techniques like the Synthetic Minority Over-sampling Technique (SMOTE) or utilizing cost-sensitive learning algorithms could help the models better recognize the minority class (enrolled students) without generating as many false positives and improve model performance. This is because SMOTE creates new synthetic data points by extracting information between existing minority points, increasing the variability in the data. By doing this, model generalization will increase as the decision boundary will be more balanced.

Furthermore, there is potential for advanced feature engineering. Incorporating additional data points such as student engagement metrics (e.g., campus visit attendance, communication response rates) or more detailed financial aid details could provide the classifiers with much deeper predictive context. One could also look into NC State specific features, such as athletics, as some students in the incoming class look to play sports for certain schools as well. While some may say that this may overcomplicate the existing models, it is worth looking at to customize these predictions more to the exact NC State school system it is trying to target.

Finally, exploring advanced gradient boosting frameworks like XGBoost, paired with model explainability tools like SHAP (SHapley Additive exPlanations), could help bridge the gap we observed between achieving high predictive performance and maintaining the actionable insights required by non-technical admissions stakeholders.

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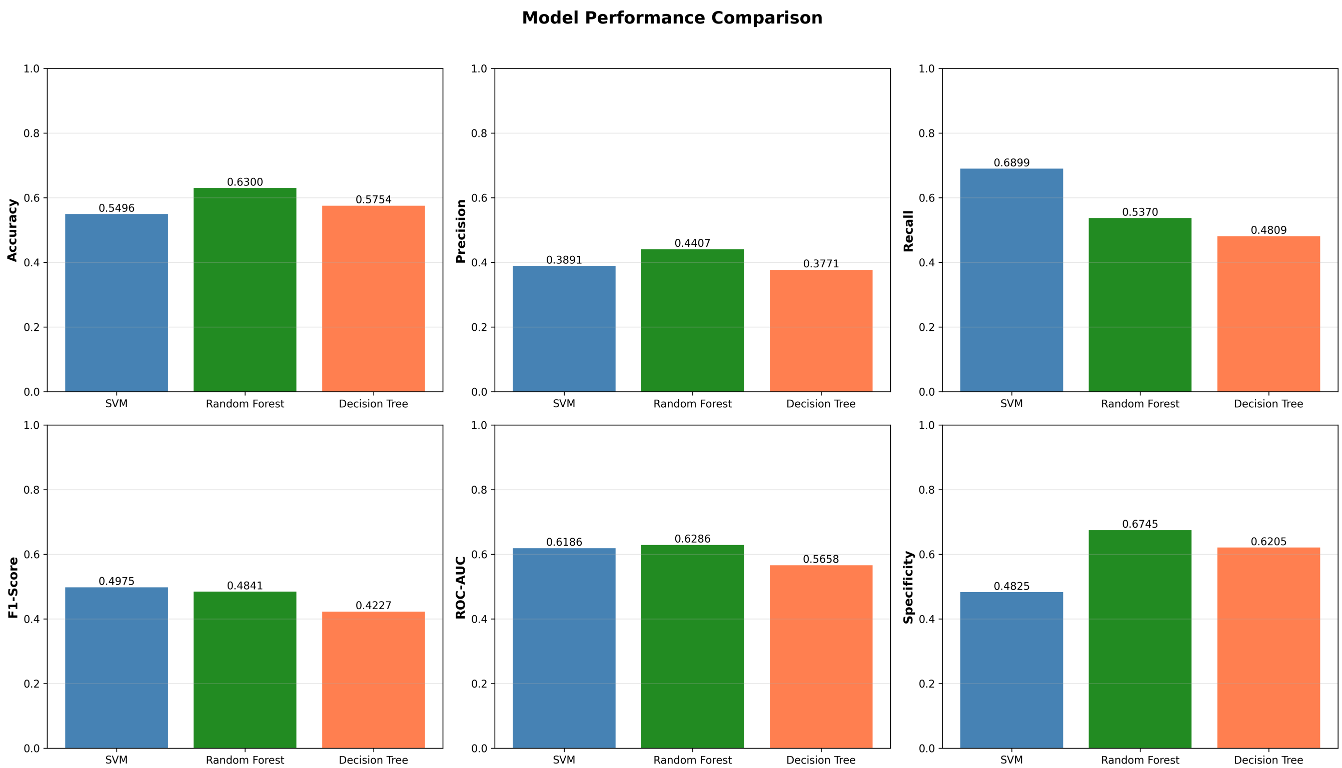

Figure 4: Performance metric comparison between the SVM, Random Forest, and Decision Tree models.

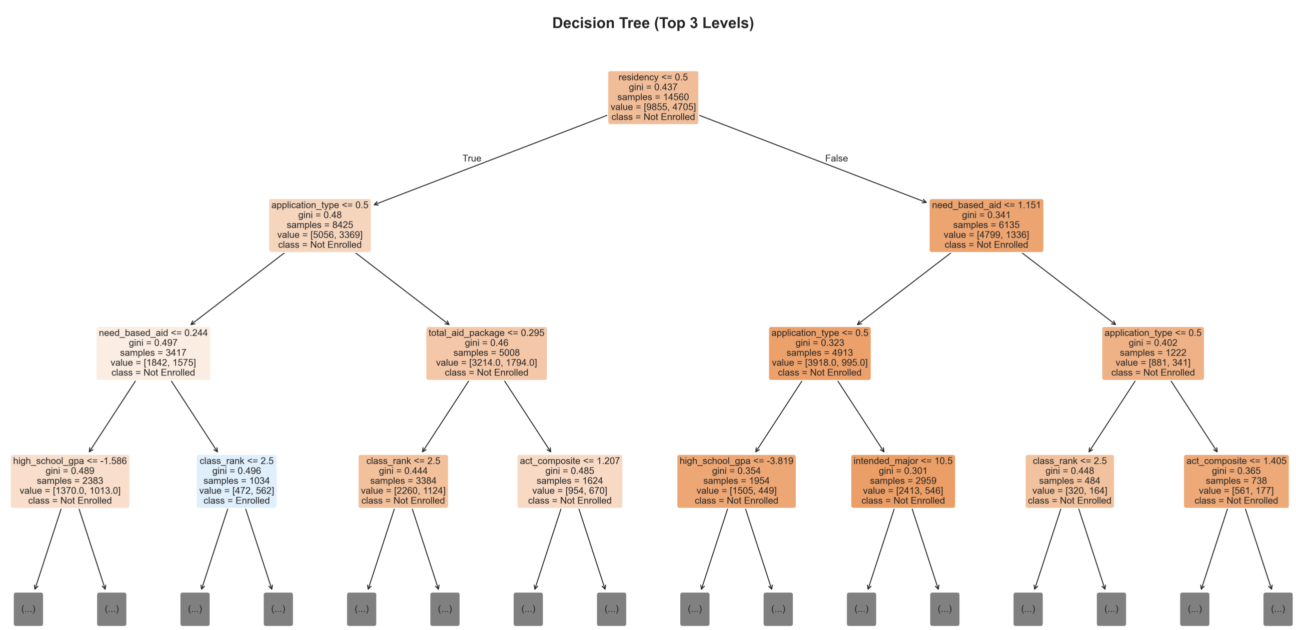

Figure 5: Visualization of Decision Tree splits, displaying the top 3 levels.